

Weekly Progress Report

Submission 1

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Abstract

This week was devoted to building the theoretical foundation for the project and surveying the relevant literature. The applied target is chaotic time-series prediction, with electrocardiogram (ECG) signal forecasting as the motivating application. The work moved from the fundamentals of chaos theory and its predictability limits, through machine-learning approaches to forecasting chaotic systems (with a focus on reservoir computing), to evaluation methodology and an initial research direction. It closed with targeted literature gathering on data-analysis pipelines and on successful forecasting methods in adjacent fields such as numerical weather prediction.

1. Theoretical Foundations

- Conducted a literature review framing the problem statement, and built up an understanding of the core theory: chaos theory, dynamical systems, and the Lyapunov exponent as a measure of sensitivity to initial conditions.
- Studied the central challenge of chaos theory, namely the exponential divergence of nearby trajectories, which imposes a fundamental predictability horizon, and how machine learning can be brought to bear on it. Related this to the target application of ECG signal forecasting, where the recorded dynamics are quasi-periodic and can exhibit chaotic behaviour.

2. Machine Learning for Chaotic Systems

- Examined the limitations of current machine-learning approaches when applied to chaotic systems, and the reasons these limitations arise.
- Studied reservoir computing (echo state networks) as a leading approach for learning chaotic dynamics, and reviewed the primary literature on its limitations.
- Surveyed how the performance of machine-learning methods on chaotic systems is evaluated, identifying the following regimes:
 - **Immediate (one-step) prediction:** forecasting the next value of the signal. This is the simplest regime, though it is not error-free: the small errors made here are what accumulate during autoregressive rollout and drive the failure of long-horizon prediction.
 - **Valid prediction time:** how long the forecast stays close to the true trajectory before diverging, typically measured in Lyapunov times. This is the most common single metric in the reservoir-computing literature and bridges short-term accuracy and long-term behaviour.
 - **Long-term qualitative agreement:** whether the forecast stays on the correct attractor and reproduces the overall pattern of the dynamics, rather than the exact

values. For ECG, this is the clinically meaningful question, such as whether the forecast trends in the same direction as a patient entering a cardiac event.

- **Long-term pointwise accuracy:** exact value-by-value agreement over a long horizon. This is extremely difficult because of sensitivity to initial conditions: a positive Lyapunov exponent causes small errors to grow exponentially, which sets a hard predictability limit.

3. Initial Research Direction

- Formed an initial hypothesis on how the underlying prediction problem might be improved, and on the potential role of the effective (finite-time) Lyapunov exponent in characterising and improving forecast performance. This remains an early idea to be developed further.

4. Literature Gathering

- Collected papers on data analysis in order to design robust and efficient pipelines for processing the project data.
- Collected papers directly addressing the target problem and the ways the existing literature has approached it.
- Began surveying adjacent fields, notably numerical weather prediction, to identify transferable methods. Successful examples reviewed include graph neural network models (such as GraphCast) and 3D transformer architectures (such as Pangu-Weather’s 3D Earth-Specific Transformer).

5. Planned for Next Week

- Read and summarise the gathered papers in depth, with written feedback on each.
- Develop the initial research direction into a more concrete proposal.